

Introduction to Flutter

AI Tutor - Flutter - Lesson 1

Introduction

Flutter is an open-source UI framework created by Google for building natively compiled applications for mobile, web, and desktop from a single codebase, using the Dart programming...

Explanation

Flutter is an open-source UI framework created by Google for building natively compiled applications for mobile, web, and desktop from a single codebase, using the Dart programming language. In Flutter, everything you see on screen is a widget: buttons, text, layout containers, even the app itself. Widgets combine to build a UI, and Flutter's 'hot reload' feature lets developers see code changes instantly without restarting the app, making UI development fast and iterative.

Key Points

- Flutter apps are written in the Dart programming language.
- Everything visible in a Flutter app is a widget.
- Hot reload shows code changes instantly without restarting the app.
- One codebase can target Android, iOS, web, and desktop.

Examples

- `Text('Hello, Flutter!')` displays a text widget.
- A `Column` widget arranges its children vertically.
- An `ElevatedButton` widget displays a clickable button.

Summary

Flutter lets developers build native apps for multiple platforms from one Dart codebase, composing everything from widgets, with hot reload for fast iteration.

Practice Questions

- 1 What programming language does Flutter use?
- 2 What is a widget in Flutter?
- 3 What does 'hot reload' let a developer do?

Quiz

Q1. What programming language is used to write Flutter apps?

a) Java | b) Swift | c) Dart | d) Kotlin

Q2. In Flutter, buttons, text, and layouts are all examples of:

a) Packages | b) Widgets | c) Plugins | d) Themes

Q3. What feature lets you see code changes instantly without restarting the app?

a) Hot reload | b) Cold start | c) Live share | d) Fast build